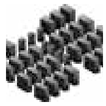
New hard metal
A super hard, new metal has been made by melting down gold and titanium
dgit.in/hardmet



Supercomputer? Nope
Scientists have developed a PC nine times faster than a supercomputer
dgit.in/SuperPc

MR. BHAVIN SHAH
FOUNDER & EVP SALES & MARKETING AT VOLANSYS

“Industries such as Healthcare, Transportation, Retail, Oil & Gas, Aviation, etc will definitely leverage IoT-ready platforms and frameworks to expand”

entrepreneurs, accommodate failures and invigorate the research sector in IoT.

The most exciting thing about this field right now is the sheer number of possibilities. When computers were invented, no one knew that someday the entire world would work on them. The same can be said for the Internet of Things. Even though there are quite a few defined roles and positions now, all that might change pretty soon since standards in this domain are still being created. “Various industries such as Healthcare, Transportation, Retail, Oil & Gas, Aviation, etc will definitely leverage IoT-ready platforms and frameworks to expand their business and connect with their audience. This is a huge potential for various IT companies to realign their offering which means deploying more workforce along with the need for special resources. Further, this is not limited to just software companies but also include semiconductor and hardware platform companies”, says Mr. Bhavin Shah, Founder & EVP Sales & Marketing at Volansys.

While it is essential to develop expertise in a specialisation among the ones mentioned above, it would also be important to figure out which industry you would be working in. There’s no point being skilled if you are not satisfied with what you would be working with. As of now, multiple sectors in India are showing great demand for IoT. Here are a few significant ones –

Manufacturing – The manufacturing industry is generating one of the highest demands for IoT implementations in their processes. Connected factories and plants are more efficient, productive and overall smarter compared to their unconnected counterparts. Data shows that production can actually double in some cases. The manufacturing industry is using historical as well as real time data to predict and avoid any factors that might affect production adversely. With sensors being implemented at every step of the manufacturing process, the actual status of factory floors is available for decision makers. Take the example of Sine Wave, a company that specializes in creating technological solutions for businesses. They created an IoT system to be used in mining operations for safety purposes. This browser based software platform is connected to devices being used within the mines and updates the status of operations in real time, allowing much better hazard resolution. Some of the leading global brands to have implemented IoT into their manufacturing processes successfully are Siemens, General Electric, Harley Davidson and Cisco.

Healthcare – Digitizing and streamlining healthcare data can have significant impact on availability of quality healthcare. On top of that, implementing the correct sensors could bring down patient monitoring cost and effort to a minimum, allowing hospitals to spend their resources in other areas that need them. A patient can actually stay at home with no loss in the efficiency of monitoring by doctors and nurses. One of the biggest concerns here is security, which can be solved by keeping the transmitted patient data anonymous and storing it locally at the hospital. A relative expertise in the medical domain would be highly sought after in the field of medical IoT. A little more advanced aspect of medical IoT is the development of smart medicines to track medication regimes, treatment effectiveness and disease symptoms from within the body.

Transportation – Since the advent of GPS, the biggest issue faced in implementing traditional IT solution in transportation systems is their mobile nature. This is tackled ideally by smaller sizes of IoT devices and their low power requirements which allow for highly portable batteries. Fleet management is already being improved in leaps and bounds by implementing IoT in bus fleets. The logistics aspect of transportation is also using IoT to improve end to end visibility. And with consumer cars being made smart, connected cars are fair playground for IoT implementations – for things like maintenance and performance management.

Home Automation – Monitoring the performance of your appliances to remotely controlling the same, home is actually the perfect use case for IoT implementation. With the home automation industry itself showing a boom, demand for skilled IoT professionals in this segment is bound to increase soon.

Consumer products – There are a large number of gadgets being made for direct consumer use. “I think consumer as a segment is what’s picking up because the products are more out there. For example, look at all these health and fitness bands you see. They are all in a sense IoT devices; have lots of sensors,

DR. KAUSTUBH NANDE
COUNTRY MARKETING HEAD ANSYS INDIA

“I think you can segment IoT skills mainly into core engineering skills that could span across a number of technical areas”